Structure-Adaptive Oriented Object Detection Network for Remote Sensing Images

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***Abstract*— Nowadays, high-resolution remote sensing images provide rich data sources and deep learning models show powerful feature representation capability for remote sensing object detection. However, due to the complex object structure as well as the changeable rotation angle, how to efficiently estimate the oriented bounding box regarding the accurate location of objects is still an open issue. Focused on this, a new one-stage structure-adaptive oriented object detection (SOOD) network is proposed, in this article. First, we designed a new rotation angle encoder (RAE), where an angle coordinate system is adopted and periodic angle correction is conducted. Different from the traditional longe-edge definition for angle estimation, the RAE can mitigate boundary discontinuity and square-like problems. Then, structure-adaptive label assignment (SALA) and confidence estimation (SACE) are introduced, to locate the position of objects more accurately. On the one hand, the anchor box determines the label assignment according to the affiliation relationship between the center point and the object's inner ellipse boundary. By constraining the ellipse boundary and employing non-parametric label assignment, high-quality anchor boxes are initially selected, and low-quality anchor boxes are suppressed. On the other hand, the integration of intersection over union (IoU) prediction and uncertainty prediction constructs a quality evaluation function to guide. In this manner, this function dynamically evaluates the localization and classification ability of each prediction box. Extensive experiments on publicly available datasets such as DOTA1.0, DOTA1.5, DIOR, and MAR20 demonstrate the effectiveness of the proposed model. The source code will be available at https://github.com/fan609/SOOD.**

***Index Terms*— Oriented object detection (OOD), remote sensing image, rotation angle encoder (RAE), structure-adaptive confidence estimation (SACE), structure-adaptive label assignment (SALA).**

I. Introduction

ORIENTED object detection (OOD) is one of the most essential and challenging tasks in the remote sensing domain, which intends to provide precise localization and category of rotating targets [1], [2], [3], [4], [5], [6], [7], [8]. However, detecting densely distributed objects in complex scenes makes accurate OOD a difficult task. Moreover, objects with varying rotation degrees and structures further affect the OOD performance negatively. Thus, developing a more effective and accurate OOD method has become a hot topic in recent years [9], [10], [11], [12].

For OOD, there are two main approaches based on deep learning techniques, i,e., two-stage and one-stage methods. The two-stage method follows a coarse-to-fine process [13], [14], [15], [16], [17], [18], [19], which first roughly predicts the location of foreground objects to get region proposals, and then localize and classify the proposals. Among these papers, most works adopt CNN to archive the OOD [16], [17], [18], [19], Then, some other methods aim at revising network structures to improve detection performance, e.g., multiscale network [16], Transformer base network [15]. In addition, works in [17], [18], and [19] focused on the design of orientation estimation, to more accurately estimate the rotated angle of objects. However, the two-stage method has the disadvantages of slow inference speed and large model capacity. Differently, the one-stage detection method performs dense prediction on the image to get the location and category of the object at the same time. Classically, you only look once (YOLO) series [20], [21], [22], single-shot multibox detector (SSD) [23], and fully convolutional one-stage (FCOS) [24] are well-known models for one-stage OOD. Based on them, some modified and revised rotated object detection methods [25], [26], [27], [28], [29] are also conducted to improve detection performance further. In contrast, the one-stage manner has the advantages of higher real-time performance and small model capacity compared to the two-stage manner, which is of high value in practical applications. As a result, there is currently a strong research focus on developing new rotated object detectors of one-stage.

In the one-stage OOD methods, there is an initial need for orientation representation to characterize the position and rotation angle of objects. At present, the rotation object generally is represented by five parameters $\{x, y, w, h, \theta\}$, where (x, y) denotes the center of the target. (w, h) represents the size of the rotation target, and θ denotes the target orientation angle. However, the five-parameter method suffers from boundary discontinuity and square-like problems. As shown in Fig. 1, the ground truth (GT) of the rectangular object should be 60°, while the prediction is −120°. When the network is training, the difference in angle between the prediction and the GT is 180°, causing a surge in angle loss and resulting in network model learning confusion. This phenomenon sometimes occurs due to the angle period of the

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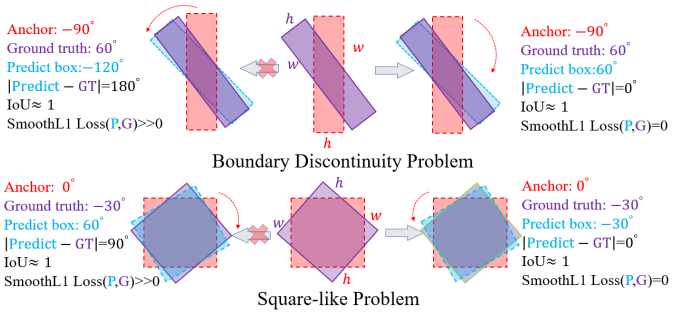


Fig. 1. Boundary discontinuity problems and square-like problems in the long-edge representation.

rectangular box is π. Another example is also given in Fig. 1, where the angle bias also appears in the square-like object prediction. Focused on this, some researchers have proposed a quadrilateral bounding box (QBB) [30] representation method based on the eight parameters [31]. It represents the four corners of the object, and the four diagonals are optimized using the smooth L1 loss. Nevertheless, the QBB method still has boundary discontinuity caused by the ordering of corner points. In contrast, Yi et al. [32] adopted the anchor-free [33] concept to define the target regression as the regression of four direction vectors. By dividing the four quadrants, the direction vectors are labeled. However, this method still encounters boundary confusion when predicting horizontal boxes, leading to boundary discontinuity.

Moreover, how to accurately predict the location and structure (e.g., shape and size) of objects is another difficulty. Focused on this, some works proposed effective label assignment strategies for distinguishing between positive anchors and negative anchors, where positive anchors represent the anchor boxes assigned to the object's foreground region and negative anchors represent the anchor boxes assigned to the background. For example, some studies [27], [34] employed an IoU threshold to determine positive and negative samples. However, the IoU threshold needs to be manually adjusted for different datasets, which lacks a good generalization. To address the issue of manually setting the threshold issue, Zhang et al. [35] proposed to dynamically calculate the IoU threshold according to the statistical characteristics of the target category, which still fails to consider the specificity of each object. Based on the label assignment results, confidence estimation is one key step that is responsible for scoring the prediction boxes in terms of position and structure. Most of the current works [25], [26], [27], [28], [29] take the classification score as the confidence estimation result, which lacks an effective measure of structure matching degrees between objects and predicted boxes. Jiang et al. [36] take IoU scores as a complement to the target structure information, but a hyperparameter tradeoff is needed to be set between the classification information and the localization information. The aforementioned confidence estimation methods suffer from inconsistency between confidence estimation and object structure estimation.

Based on the above analysis, the main bottlenecks in realizing high-precision OOD arise from two aspects. On the one hand, the boundary discontinuity and square-like problem are caused by the variable object angle representation. According to the limitation of the angle representation, we design the angle coordinate system and correct the angle to alleviate the boundary discontinuity and the square-like problems. On the other hand, the estimation bias of object localization and size is generated by the complex object structure. The estimation bias of object localization and size mainly stems from the mismatch of oriented objects and anchors in the label assignment. Given the complex structure of the object, we use the internal ellipse constraint and adaptive selection of candidate anchors to filter the set of best-matching candidate anchors. To screen the prediction boxes with the optimal localization and classification ability during the non-maximal suppression procedure, a confidence evaluation function with the ability to dynamically evaluate localization and classification is adopted to reasonably evaluate each prediction box. To mitigate these issues, we propose a structure-adaptive OOD network (SOOD) in this work. The contributions of this work can be summarized as follows:

1) To better characterize the object orientation representation, a new rotation angle encoder (RAE) is proposed. Different from the traditional long-edge representation, the RAE characterizes object orientation and conducts a periodic angle correction in an angle coordinate system. In this manner, it effectively alleviates loss surges caused by boundary discontinuity and square-like problems.
2) Focused on the mismatch between anchors and referred rotated boxes of objects, a novel structure-adaptive label assignment (SALA) strategy is designed. Here, the SALA introduces an internal ellipse constraint on anchors and adaptively selects candidate anchor boxes according to object structure. In this way, more accurate label assignments can be achieved, where negative anchors are eliminated and positive ones are selected effectively.
3) To realize the consistency of the confidence estimation with the object structure, the structure-adaptive confidence estimation (SACE) method is proposed. In brief, the SACE introduces an uncertainty measure to assess the location/structure estimation results of the object box, which will guide the dynamical balance of the classification and localization predictions by joint loss optimization. As a result, OOD results can be improved.

The remainder of this article is organized as follows. In Section II presents the details of our proposed method. Section III describes and discusses the experimental results. Finally, conclusions are drawn in Section IV.

II. Proposed Approach

In this section, details of the proposed SOOD method will be described, where the flowchart is shown in Fig. 2. The gray component is the subject of the network, the SOOD uses ConvNeXt-T [37] as the backbone network (which was pretrained on the ImageNet-22K dataset), while the feature pyramid [38] and path aggregation network [39] (FPN-PAN) as the feature aggregation module. The P-Head is mainly divided into localization and classification branches. First, the

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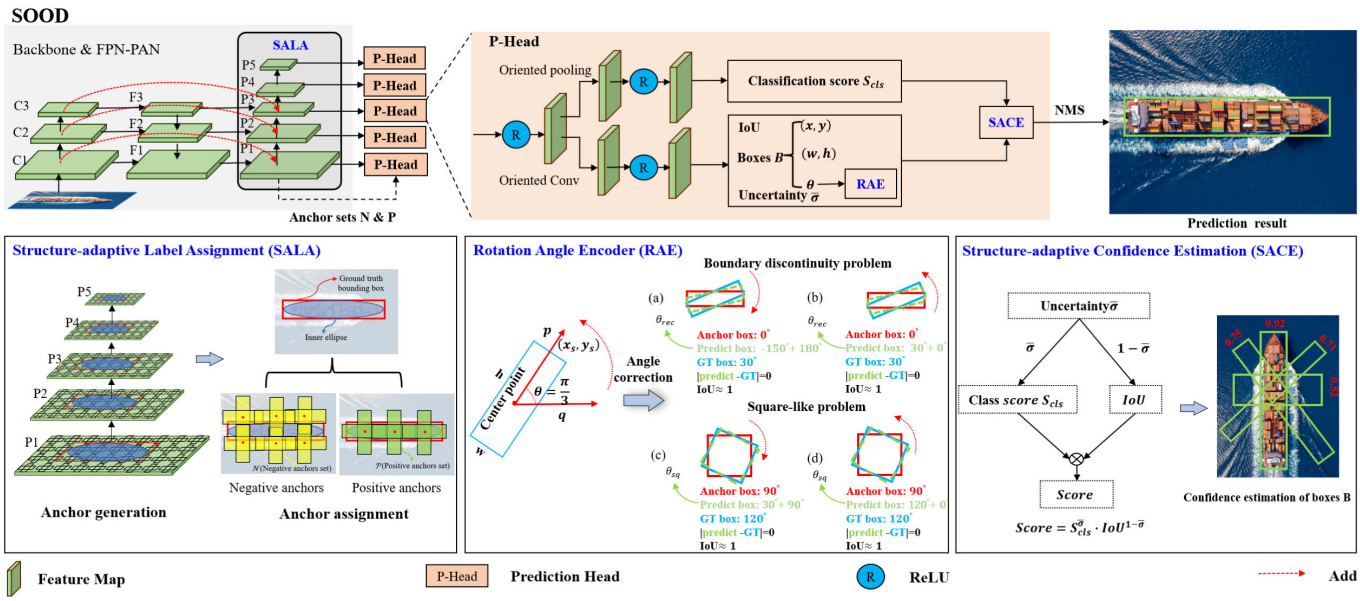


Fig. 2. Architecture of the proposed SOOD method. The RAE is used to encode angles during training and inference. The SALA performs anchor box assignment at the multiscale feature layer. The SACE reasonably evaluates the prediction box score in the prediction head. NMS denotes non-maximal value suppression operation.

multiscale features of FPN-PAN are extracted by oriented convolution for orientation-sensitive features. Then, the IoU, uncertainty, and prediction boxes are predicted by the four conventional convolutions of the localization branch. Finally, the classification branch performs directed pooling and four conventional convolutions on the orientation-sensitive features to predict the target category. The RAE is a rotation angle encoder that mitigates boundary discontinuities and square-like problems arising during network training. It is responsible for converting and correcting the angles predicted by the network to the angles of the RAE's angle coordinate system and taking part in the training and inference of the network model. The SALA is a structure-adaptive label assignment strategy that helps to achieve structure consistency between anchors and referred rotated boxes of objects. The SACE is structure-adaptive confidence estimation that mines reliable location and structure information for each predicted box. Which dynamically assesses the classification and localization ability of the box. Details of the proposed method are given below.

A. Rotation Angle Encoder

To mitigate the boundary discontinuity and square-like problems in the long-edge representation, we proposed the RAE that periodically corrects for rotation angles exceeding the defined range. As shown in Fig. 2, the input of this module is the angle θ predicted by the prediction head, and its output is the angle based on the new angle coordinate system. The RAE is primarily responsible for angle conversion and rectification in the training and inference phases of the network model. When the object's angle regresses to the boundary, two directions of regression can be employed for predicting the angle of the bounding box. In one of the directions, the angle prediction may exceed the defined range, resulting in a noticeable increase in angle loss. Our RAE is described below.

First, the five parameters $\{x, y, w, h, \theta\}$ are transformed into four corner points. Here, the center coordinates of the target are represented by (x, y), the target size is represented by (w, h), and the orientation angle is denoted by θ. The conversion can be performed using the following formulas:

$$\mathbf{vec}_1 = \left[\frac{w}{2}\cos\theta \quad \frac{w}{2}\sin\theta\right] \tag{1}$$

$$\mathbf{vec}_2 = \left[-\frac{h}{2}\sin\theta \quad \frac{h}{2}\cos\theta\right] \tag{2}$$

$$\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix} = \begin{bmatrix} x & y \\ x & y \\ x & y \\ x & y \end{bmatrix} + \begin{bmatrix} \mathbf{vec}_1 \\ \mathbf{vec}_1 \\ -\mathbf{vec}_1 \\ -\mathbf{vec}_1 \end{bmatrix} + \begin{bmatrix} \mathbf{vec}_2 \\ -\mathbf{vec}_2 \\ -\mathbf{vec}_2 \\ \mathbf{vec}_2 \end{bmatrix}. \tag{3}$$

In (3), the coordinates of the four corners of the object are represented by (x_1, y_1), (x_2, y_2), (x_3, y_3), and (x_4, y_4), respectively. The combination of $\mathbf{vec_1}$ and $\mathbf{vec_2}$ with different signs represents the directional vectors in the global coordinate system, obtained by rotating the local coordinate system direction vectors. The four corner points are sorted and the lowest corner point coordinates are selected, establishing the orientation axis using the short side near the corner point.

Second, an angle coordinate system is established with the target's center point as the origin. As shown in Fig. 2 for the RAE, the angle coordinate system consists of two red arrows, where the orientation axis points to the shorter edge and the horizontal axis to the right. The angle between two axes is calculated as the object's orientation angle using the formula

$$\cos\theta_r = \frac{\mathbf{p}\cdot\mathbf{q}}{|\mathbf{p}||\mathbf{q}|} \tag{4}$$

where $\mathbf{p}$ represents the vector of the orientation coordinate axis and $\mathbf{q}$ represents the vector of the horizontal coordinate axis. The loss varies depending on the direction of the object box

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regression. The vector of directions $\mathbf{q}$ is $(1, 0)$. The direction vector $\mathbf{p}$ is $(x_s - x, y_s - y)$. (x_s, y_s) is the center point of the short axis, which is calculated from the coordinates of the corner points of the rotated box. θ_r is the angle of the target in the new coordinate system.

Finally, periodic angle correction is applied to boundary discontinuity and square-like problems. As shown in Fig. 2(a), if the predicted angle of the rectangular object's bounding box exceeds the boundary and is negative, π is added to the angle to confine it within the range of $[0, \pi)$. It can be seen from Fig. 2(b) that the regression direction of the rectangle prediction box is opposite to that in Fig. 2(a). At this time, the angle of the prediction box is within the angle range $[0, \pi)$, which will not cause a sharp increase in the angle loss. Therefore, no angular compensation is required in this case. For square-like targets, as is exhibited in Fig. 2(c), if the y-coordinates of the corner points above the center point are equal, the angle of the square-like target is set to $(\pi/2)$. Considering that the period of a square-like target is $(\pi/2)$, negative predicted angles for square-like targets are compensated by adding π. Similarly, for predicted angles within the range of $[0, (\pi/2))$, $(\pi/2)$ is also added. Fig. 2(d) shows that the regression direction of the square prediction box is opposite to that of Fig. 2(c), and the angle is within the angular range $[(\pi/2), \pi)$ of the square object, so no angle compensation is required. Angles of rectangles are encoded in the range $[0, \pi)$ and squares are encoded in $[(\pi/2), \pi)$. The specific formulas for these corrections are as follows:

$$\theta_{\text{rec}} = \begin{cases} \theta_r + \pi, & \text{if } \theta_r < 0 \\ \theta_r, & \text{otherwise} \end{cases} \tag{5}$$

$$\theta_{\text{sq}} = \begin{cases} \theta_r + \pi, & \text{if } \theta_r < 0 \\ \theta_r + \dfrac{\pi}{2}, & \text{if } 0 \le \theta_r < \dfrac{\pi}{2} \\ \theta_r, & \text{otherwise.} \end{cases} \tag{6}$$

In (5), the corrected predicted angle for rectangle targets is denoted as θ_{rec}, and in (6), the corrected prediction for square targets is represented by θ_{sq}. The corrected angle θ_{sq} or θ_{rec} are all labeled θ_{rae}.

B. Structure-Adaptive Label Assignment

To achieve consistency in the structure of anchor boxes and referred rotated boxes of objects, we proposed a SALA strategy. The inputs to SALA are anchors and GT boxes on the multiscale feature map, and the outputs are the set of positive sample anchors and negative sample anchors. Here, we select top k high-quality anchor boxes for label assignment by rotating the target structure. It is worth noting that the strategy is designed without hyperparameters, which are universal to various datasets. The processes are summarized in **Algorithm** 1.

To begin with, we employ the angle encoding by RAE to transform the image coordinate system into the rotating target coordinate system

$$\begin{bmatrix} x_r \\ y_r \end{bmatrix} = \begin{bmatrix} \cos\theta_{\text{rae}} & -\sin\theta_{\text{rae}} \\ \sin\theta_{\text{rae}} & \cos\theta_{\text{rae}} \end{bmatrix} \begin{bmatrix} x - x_c \\ y - y_c \end{bmatrix} \tag{7}$$

Algorithm 1 Structure-Adaptive Label Assignment SALA

Input:

$\mathcal{G}$ is a set of ground-truth boxes in the input image.
$\mathcal{A}$ is all the anchors.
$\emptyset$ is an empty set.
$IoU(\cdot)$ means to calculate the intersection of union between two oriented boxes.
$Distances(\cdot)$ indicates the euclidian distance calculated from the center points of the two oriented boxes.

Output:

$\mathcal{P}$ is a set of positive anchors.
$\mathcal{N}$ is a set of negative anchors.

Calculate the distance between the centers of $\mathcal{G}$ and $\mathcal{A}$ $Distances(\mathcal{G}, \mathcal{A})$
Calculate the $\mathcal{G}$ and $\mathcal{A}$ rotation IoU $IoU(\mathcal{G}, \mathcal{A})$
$\mathcal{P} \leftarrow \emptyset$
for each ground truth $g \in \mathcal{G}$ **do**
  Calculate top k by Eq.(8)
  $S \leftarrow$ Select top k anchors from $IoU(g, \mathcal{A})$
  $C \leftarrow$ Select anchors inside g's ellipse from S
  if $C = \emptyset$ **then**
    $C \leftarrow$ Select top k anchors from $Distances(g, \mathcal{A})$
  end if
  $\mathcal{P} = \mathcal{P} \bigcup C$
end for
$\mathcal{N} = \mathcal{A} - \mathcal{P}$
return $\mathcal{P}, \mathcal{N}$

where (x_c, y_c) represents the center point of the oriented target, (x, y) denotes any point in the original coordinate system, and (x_r, y_r) corresponds to the coordinate point in the rotated coordinate system. The angle θ_{rae} signifies the rotation angle of the coordinate system.

Then, an inner ellipse boundary is established based on the length and width of the target within the rotating coordinate system. We calculate the distance between the center point of the anchor box and the center point of the GT box. The number of adaptive candidate boxes (k) is determined by the size of each GT box and the sampling strides

$$k = \text{ceil}\left(\sum_{i=1}^{n} \frac{\max\{w, h\}}{s_i}\right) \tag{8}$$

where ceil$(\cdot)$ is upward rounding and n denotes the downsampling times. Here, s_i represents the downsampling multiples of the feature map, and it takes the value $[4, 8, 16, 32, 64]$. max$(\cdot)$ represents the operation of taking the maximum value. h and w denote the size of the object.

Finally, we adopt the top k anchor boxes with relatively high IoU as candidate boxes for regression. From these candidates, we select anchor boxes within the elliptical boundary as positive sample boxes. If there are no positive anchor boxes inside the ellipse boundary, the k anchor boxes closest to the center of the GT box are selected as positive anchor boxes. When an anchor box matches multiple GT boxes, we select the GT box with the largest IoU to anchor. Fig. 3 visualizes the results of structural-adaptive label assignment of anchor boxes.

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(a) (b)

Fig. 3. Visualizing the results of adaptive structural label assignment of anchor boxes during training. (a) Blue boxes denote ground truth boxes. (b) Green points represent the center of the anchor boxes inside the ellipse. The red points indicate the centers of the top k positive anchor boxes.

Fig. 3(a) shows two aircraft types, with larger and smaller wingspans, respectively. From Fig. 3(b), it can be seen that the positive anchor box centers are more concentrated on compact aircraft models while larger wing aircraft models have more centers of pseudo-positive anchor boxes.

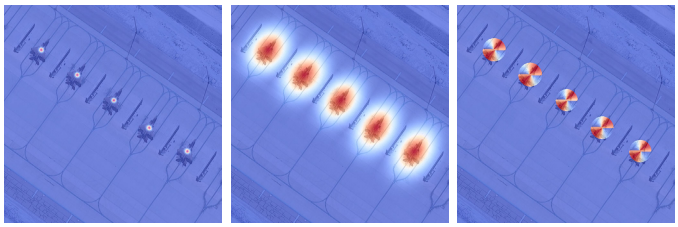
C. Structure-Adaptive Confidence Estimation

To address the issue of over-dependence of prediction box results on classification confidence, we introduce SACE. It takes the classification scores, IoU, uncertainty, and prediction boxes as inputs and performs a reasonable evaluation for each prediction box before NMS. The SACE considers the distinct characteristics of each prediction box by jointly exploiting its classification and localization capabilities. This integration enables more accurate and reliable predictions. The score of SACE is given below

$$\text{Score} = S_{\text{cls}}^{\overline{\sigma}} \cdot \text{IoU}^{1-\overline{\sigma}} \tag{9}$$

where S_{cls} and $\overline{\sigma} = \sum_{j=1}^{5} \sigma_j$ are the classification confidence score (CCS) and the mean of uncertainty prediction, respectively. σ_j denotes the uncertainty of the jth prediction variable(e.g., (x, y, w, h, θ)). IoU is the rotation intersection-over-union, which represents the location capability. As shown in Fig. 4, gaussian distributions for object center point, size, and angle are obtained by using the uncertainty as the standard deviation and the prediction as the mean. Where the red area means low uncertainty and the blue color indicates high uncertainty.

1) Loss Function: The total loss function mainly consists of location loss L_{loc}, IoU loss L_{iou}, angle loss L_{agl}, and classification loss L_{cls}. Location loss is responsible for regressing the target center point and size. IoU loss is charged with optimizing the network's IoU prediction. Angle loss is accountable for optimizing angle estimation for rectangle and square-like targets. The classification loss is for optimizing the network category prediction.



(a) (b) (c)

Fig. 4. Visualizing the results of the structural uncertainty of the objects. (a) Uncertainty of the object center point. (b) Uncertainty of the object size. (c) Uncertainty of the object orientation angle.

The location branch optimization typically employs L1 or smooth L1 loss for its five parameters. However, this optimization method assumes that variables are independent of each other despite their correlations. It is beneficial to design a loss function that accounts for the uncertainties and correlations among predictor variables. By leveraging the IoU loss constraint, the loss function can optimize the geometric relationship between variables and extract more reliable target information for regression based on uncertainty constraint. The function's mathematical expression is as follows:

$$\begin{aligned} L_{\text{loc}} &= \sum_{i=1}^{N} \delta_i \text{Smooth}_{L1}\left(\boldsymbol{D_i} \cdot \left|-\log\left(\text{IoU}_i\right)\right| \big/ \boldsymbol{\sigma_i} + \log\left(\boldsymbol{\sigma_i}\right)\right) \\ \text{s.t.,} \quad & \boldsymbol{D_i} = \left|\mathbf{t}'_i - \mathbf{t}_i\right| \end{aligned} \tag{10}$$

where $\text{Smooth}_{L1}(\cdot)$ denotes the smooth L1 loss. N represents the overall number of samples. Moreover, the notation δ_i is the indicator function that specifies the ith anchor box as a positive sample by matching the GT box. $\mathbf{t}'_\mathbf{i}$ denotes the predicted center point and size of the ith target. $\mathbf{t}_\mathbf{i}$ is the center point and size of the ith GT box. $\mathbf{D}_\mathbf{i}$ is the difference between the absolute values of $\mathbf{t}'_\mathbf{i}$ and $\mathbf{t}_\mathbf{i}$. $\boldsymbol{\sigma_i}$ indicates uncertainty prediction of the center point and size corresponding to the ith positive sample, while IoU_i denotes the rotation intersection-over-union between the ith prediction box and the GT box. Furthermore, the loss function for the IoU branch is the cross-entropy loss, which can be expressed as follows:

$$L_{\text{iou}} = \sum_{i=1}^{N} \delta_i L_{\text{ce}}\left(\text{IoU}_{\text{pred}_i},\ \text{IoU}_{\text{gt}_i}\right) \tag{11}$$

where L_{iou} expresses the loss function for the localization branch. N represents the total number of samples. L_{ce} denotes the cross-entropy loss. The function δ_i is an indicator function that shows whether the ith anchor box matches the GT box. $\text{IoU}_{\text{pred}_i}$ predicts the IoU for the ith object, and IoU_{gt_i} denotes the true IoU of the prediction box with the GT box.

To alleviate the problem that the IoU of the square-like object is not sensitive to the loss function. When the square target prediction box is rotated by an angle, the angular loss increases significantly, but its IoU remains almost unchanged. The change in the IoU indicator is related to the aspect ratio of the target. The closer the aspect ratio is to 1, the smaller the

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IoU change. For square-like objects, the insensitivity of the IoU causes severe jittering of the box boundary, which affects the stability of the box. Accordingly, we design an angular loss function that is sensitive to square-like objects

$$L_{\text{agl}} = \sum_{i=1}^{N} \delta_i \text{Smooth}_{L1}\left(T_i \cdot |-\log\left(\text{IoU}_i\right)|/\sigma_{\theta_i} + \log\left(\sigma_{\theta_i}\right)\right)$$
$$\text{s.t.,} \quad T_i = r_i \cdot \sin(4\Delta\theta_i) + \sqrt{1-r_i^2} \cdot \sin(2\Delta\theta_i) \tag{12}$$

where r_i is the aspect ratio of the ith target and its value range is (0, 1]. $\Delta\theta_i = |\theta_{\text{pred}_i} - \theta_{\text{gt}_i}|$ is the absolute difference between the predicted angle of the ith target and the GT angle. θ_{pred_i} and θ_{gt_i} are the prediction angle and the GT label of the ith target after RAE encoding. σ_{θ_i} is the angle uncertainty of the ith target. T_i represents the angular sinusoidal loss that varies with the rotating box period and size. The periodicity of the angles of the square-like and rectangular targets is integrated into the above equation to address the inconsistency of the angle loss and IoU metric on the square-like targets. When r_i is close to 1, the bounding box is close to square-like, then $\sin(4\Delta\theta_i)$ acts as an amplitude regulator for the IoU, giving the IoU a periodicity close to that of the square-like target. The same principle for rectangular targets. The regression loss function is given by

$$L_{\text{reg}} = L_{\text{loc}} + L_{\text{iou}} + L_{\text{agl}}. \tag{13}$$

The classification branch is optimized using the focal loss function, which can be expressed as follows:

$$L_{\text{cls}} = -\alpha_t(1-p_t)^{\gamma}\log\left(p_t\right)$$
$$\text{s.t.,} \quad p_t = \begin{cases} p, & \text{if } y = 1 \\ 1-p, & \text{otherwise} \end{cases} \tag{14}$$

where y denotes the label, p represents the confidence of the predicted category, p_t is an intermediate representation. a_t and γ are the hyperparameters set to 0.25 and 2, respectively. N_{pos} denotes the number of positive samples. The total loss function can be expressed as

$$L_{\text{total}} = \frac{1}{N_{\text{pos}}}\left(L_{\text{reg}} + L_{\text{cls}}\right). \tag{15}$$

III. Experiments

In this section, we describe the experiments conducted to demonstrate the proposed SOOD's effectiveness for OOD. First, we briefly describe the four datasets of oriented detection used in our experiments. Second, experimental parameter settings and evaluation metrics for different datasets are given. Third, ablation experiments are performed to verify the effectiveness of each module. Finally, we perform comparison experiments of our method with some state-of-the-art methods.

A. Datasets

Here, the four publicly available OOD datasets include DOTA-v1.0 [40], DOTA-v1.5 [40], DIOR-R [41], and MAR20 [42]. We will briefly describe these datasets below.

1) DOTA-v1.0: A large remote sensing object detection dataset. DOTA-v1.0 is a comprehensive dataset designed for remote sensing object detection. It comprises RGB and grayscale images, with sizes ranging from 800 × 800 to 20 000 × 20 000 pixels. The dataset contains objects of various scales and orientations, making it suitable for evaluating detection algorithms in challenging scenarios. DOTA-v1.0 consists of 2806 images, 15 categories, and 188 282 instances. The 15 categories include plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). The dataset offers two detection tasks: horizontal bounding box and oriented bounding box. Each instance target is annotated using a quadrangle bounding box with vertices arranged in clockwise order. The dataset provides single-scale and multiscale versions, with the single-scale version containing 21 046 training images, 5297 validation images, and 10 833 test images, while the multiscale version contains 138 883 training images, 35 462 validation images, and 71 888 test images.

2) DOTA-v1.5: An enhanced version of DOTA-v1.0 for small target detection building upon the DOTA-v1.0 dataset, DOTA-v1.5 introduces additional annotations for small objects (less than 10 pixels). This enhanced version includes a new category, container crane (CC), and expands the total number of categories to 16. DOTA-v1.5 encompasses 403 318 instance objects, providing a more comprehensive benchmark for small object detection algorithms. The division of the training, validation, and test sets for this dataset is the same as that of DOTA-v1.0.

3) DIOR-R: Object detection in optical remote sensing images. The DIOR-R dataset is specifically designed for object detection in optical remote sensing images. It consists of 23 463 images and 190 288 object instances across 20 categories. The 20 common categories are airplane (APL), airport (APO), baseball field (BF), basketball court (BC), BR, chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), golf field (GF), GTF, HA, overpass (OP), SH, stadium (STA), storage tank (STO), TC, train station (TS), vehicle (VE), and windmill (WM). The image size in DIOR-R is fixed at 800 × 800 pixels, with varying resolutions ranging from 0.5 to 30 m. The dataset captures different seasons and weather conditions, resulting in diverse imaging characteristics. DIOR-R serves as a valuable resource for evaluating object detection algorithms in the context of optical remote sensing. The dataset contains 11 725 images in the training set and validation set and 11 738 images in the test set.

4) MAR20: A large dataset for military aircraft object recognition in remote sensing images. MAR20 is a comprehensive dataset focusing on military aircraft object recognition in remote sensing images. It comprises 3842 images, covering 20 different types of military aircraft. The 20 types of military aircraft consisting of SU-35, C-130, C-17, C-5, F-16, TU-160, E-3, B-52, P-3C, B-1B, E-8, TU-22, F-15, KC-135, F-22, FA-18, TU-95, KC-10, SU-34, and SU-24, which are denoted

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TABLE I
EXPERIMENTAL PARAMETER SETTINGS ON FOUR DATASETS

Settings	DOTA-v1.0	DOTA-v1.5	DIOR-R	MAR20
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning rate	$1e^{-4}$	$1e^{-4}$	$1e^{-4}$	$1e^{-4}$
Weight decay	0.05	0.05	0.05	0.05
Epoch decay	(24,33)	(24,33)	(24,33)	(48,66)
Epochs	36	36	36	72
Batch size	2	2	4	8
Image size	1024× 1024	1024× 1024	800× 800	800× 800

TABLE II
ABLATIVE EXPERIMENTS AND EVALUATION OF THE PROPOSED SOOD METHOD ON THE MAR20 DATASET

Methods	RAE(a)	Uncertainty(u)	SACE(c)	SALA(l)	mAP
Baseline					82.97
SOOD(a)	✓				83.49
SOOD(a+u)	✓	✓			84.89
SOOD(a+u+c)	✓	✓	✓		85.05
SOOD (a+u+c+l)	✓	✓	✓	✓	**85.77**

by A1-A20. The dataset provides annotations for both horizontal bounding boxes and oriented bounding boxes. The image size in MAR20 is fixed at 800 × 800 pixels. It is worth noting that different types of aircraft in MAR20 exhibit high similarity. The dataset includes a total of 1331 images for training and validation purposes and an additional 2511 images for testing.

B. Experimental Setting

We utilize ConvNeXt-T as well as the feature pyramid and path aggregation network as the backbone network and combine the features of both networks. All experiments are conducted on a single NVIDIA GeForce RTX 3090 with 24GB memory. The experimental parameter settings for different datasets are shown in Table I. The multiscale version of DOTA-v1.0 and DOTA-v1.5 adjust the image to 0.5, 1, and 1.5 times the original image, and the images are cropped into 1024 × 1024 patches with a stride of 500.

Our model is based on the MMRotate [43] library, an open-source OOD toolbox based on PyTorch. The mAP50 of the PASCAL VOC 2007 metric is utilized as the evaluation metric for the oriented detector, and AP is adopted to verify the accuracy of each category during verification.

C. Ablation Study

In the proposed SOOD method, there are three main contribution modules, including the RAE, SALA, and SACE. Among them, the RAE can alleviate boundary discontinuity and square-like problems. The SALA attempts to solve the structural inconsistency between the anchor and the rotated target by an internal ellipse. The SACE dynamically evaluates the localization and classification capabilities of each prediction box to achieve structural self-adaptation. To demonstrate the effect of these modules, we conduct an ablation study here. Our baseline model uses ConvNeXt-T as the backbone, with multilayer and cross-layer aggregation via FPN-PAN. Direct coding is chosen for angle coding. Max IoU assignment is used for label assignment. Classification scores are used as confidence estimates. The SOOD is the abbreviation of the proposed model. "a" refers to the RAE angle code. "u" denotes uncertainty loss. "c" is the SACE estimation function. "l" is the SALA strategy. Table II presents the results of all ablation experiments. To analyze the effectiveness of each module, the effects of the above ablation experiments on each category are discussed in detail. Table III shows detailed metrics for each ablation module on 20 categories of aircraft.

1) Effectiveness of RAE: We evaluate the effectiveness of our proposed RAE on the MAR20 dataset. From the results in Table II, it can be seen that the model using RAE obtained 83.49% mAP, which is 0.52% higher than the baseline model. Table III shows the effect of the RAE module on each category, with the A11 category showing the largest improvement of 9.63% points over the baseline model. The A11 is an E-8 Joint surveillance target attack radar system (Joint STARS) with a 46.6-m fuselage and 44.4-m wingspan, which is closer to a square-like shape in the remote sensing image. The RAE has a better ability to handle square-like angle estimation. The RAE module improves by 4.07% on the A5 category compared to the baseline model. There is a degradation in accuracy in the A12 category. The A12 category is the TU-22 bomber, which has a feature that the wings can be extended and retracted according to the actual situation so that the external box changes significantly. The MAR20 dataset also has more such scenarios. Structural instability could potentially cause the A12 category recognition accuracy to decrease.

2) Effectiveness of SACE: As presented in Table II, the integration of SACE into the SOOD(a+u+c) model yields a notable improvement of 2.08% in comparison to the baseline model. The SOOD(a+u) model demonstrates a 1.40% improvement compared to the SOOD(a) model, with increases in categories such as A12 and A20 that exhibit high similarity. The SOOD(a+u+c) model exhibits improvements in categories with high similarity, such as A13 and A15, compared to the SOOD(a+u) model. These categories primarily focus on fighter aircraft. Thus, uncertainty loss can mine reliable information about the target structure and make it more discriminating. Structural adaptive confidence estimation enables the selection of more accurate prediction boxes based on reliable information about the target structure. As shown in Fig. 5, the SACE can dynamically assess its classification and localization capabilities, thus allowing for partial adjustment of the box angle.

3) Effectiveness of SALA: As depicted in Table II, the SOOD(a+u+c+l) model exhibits a 0.72% increase in mAP compared to the SOOD(a+u+c) model. Additionally, Table III highlights the enhancement of the introduced SALA strategy, particularly for compact aircraft models like A12, A13, A16, and A20. However, it is important to note a decrease in accuracy for models with larger wingspans, such as A2 and A5. The internal ellipse boundary of the SALA strategy can better constrain the anchor boxes of the compact structural object, which selects better quality positive anchor boxes through its adaptive filter mechanism. However, many pseudo-positive anchor boxes are generated on categories with large wingspans, thus causing a certain decrease in precision.

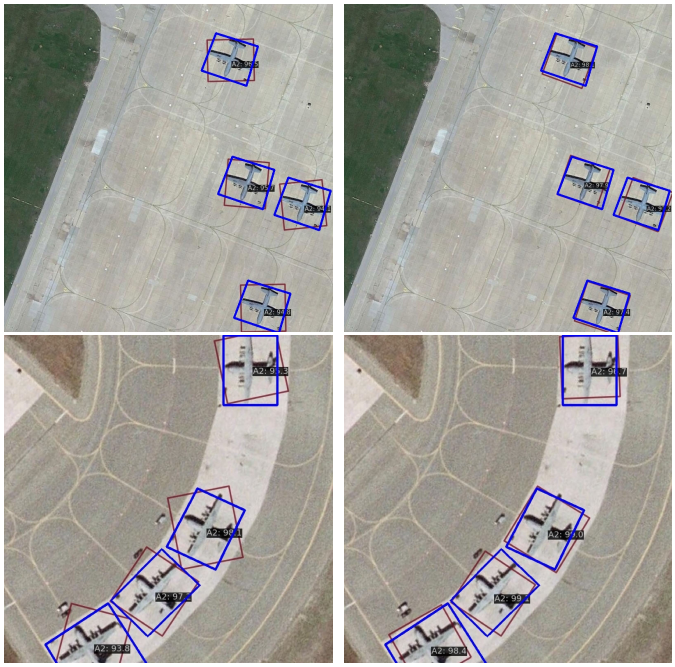
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TABLE III
MORE DETAILED ABLATION EXPERIMENTS ON MAR20 DATASET

Methods	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14	A15	A16	A17	A18	A19	A20	mAP
Baseline	85.72	89.04	87.41	83.82	73.26	90.33	89.48	88.54	90.57	90.91	77.64	83.65	65.59	85.03	62.86	81.53	90.36	84.07	83.38	76.21	82.97
SOOD(a)	88.73	**89.44**	87.45	85.11	77.33	**90.82**	87.99	89.50	90.13	90.83	**87.27**	75.62	64.07	88.97	64.87	83.82	90.42	80.01	84.37	73.05	83.49
SOOD(a+u)	**89.37**	81.71	**88.42**	**85.15**	69.86	90.43	**90.21**	88.25	**90.68**	90.91	86.41	**90.14**	68.80	**89.37**	68.69	86.56	90.39	81.37	**86.68**	84.40	84.89
SOOD(a+u+c)	87.14	84.90	86.13	82.87	63.88	90.64	85.93	**89.69**	89.44	**90.96**	85.92	88.48	73.27	87.27	77.68	86.84	90.47	**89.63**	85.72	84.17	85.05
SOOD (a+u+c+l)	86.58	85.54	86.46	79.52	**79.20**	89.77	88.81	88.66	90.17	90.73	82.67	89.78	**75.35**	88.02	80.10	88.06	90.63	84.76	84.27	**86.34**	**85.77**

TABLE IV
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET

Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
two-stage																
ROI Transformer [12]	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
SCRDet [17]	**89.98**	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	86.86	65.02	66.68	66.25	68.24	65.21	72.61
Gliding Vertex [30]	89.64	**85.00**	52.6	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
CADNet [44]	87.80	82.40	49.40	73.50	71.10	63.50	76.60	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
AOPG [45]	89.14	82.74	51.87	69.28	77.65	82.42	88.08	90.89	86.26	85.13	60.60	66.30	74.05	67.76	58.77	75.39
MFIAR-Net [46]	89.62	84.03	52.41	70.03	70.13	67.64	77.81	90.85	85.40	86.22	63.21	64.14	68.31	70.21	62.11	73.49
one-stage																
RSDet [26]	89.80	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63.90	65.60	67.20	68.00	72.20
DRN [47]	88.91	80.22	43.53	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
CSL [48]	89.34	79.68	40.86	69.94	77.69	62.04	77.49	90.87	82.88	82.07	59.97	65.30	53.56	64.04	46.61	69.49
FCOSF [49]	88.67	81.67	50.31	66.68	80.58	81.08	87.58	90.91	86.17	84.95	57.40	65.01	69.82	71.33	55.51	74.51
S^2ANet [50]	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.25	69.13	57.94	74.12
RTMDet-R-tiny [51]	89.21	80.03	47.88	69.73	**82.05**	83.33	88.63	**90.91**	86.31	86.85	59.94	62.30	74.23	71.97	57.03	75.36
SOOD	89.20	84.54	51.50	72.75	74.78	77.70	87.77	90.88	85.12	85.04	65.76	65.63	73.27	71.03	57.95	75.53
SOOD (Multiscale)	88.30	84.78	**57.88**	**80.78**	79.97	**84.19**	**88.74**	90.84	85.95	**87.56**	**69.15**	**69.13**	**77.32**	**79.22**	**74.04**	**79.86**



(a) (b)

Fig. 5. Comparison of the effects of (a) CCS and (b) SACE. Blue boxes denote the ground-truth boxes and red boxes represent prediction boxes.

D. Comparison With the State-of-the-Art

To evaluate the effectiveness of SOOD, we compare the SOOD with OOD methods, including one-stage and two-stage methods based on deep learning. The experimental parameters for the comparison methods were set according to the default settings of the relevant papers.

1) Results on DOTA-v1.0: Table IV compares the SOOD model from the one-stage and two-stage perspectives, respectively. In one stage, the SOOD is superior to other models in the categories of BD, BR, GTF, SBF, and RA. This shows that the model is more effective than other one-stage models in dealing with larger targets on the ground. In the two-stage detector, this model is better than the other two-stage models in the LV, SBF, and SP categories. The SOOD model achieves two-stage accuracy with the ConvNeXt tiny version of the backbone, which is 2.29% and 0.14% higher than SCRDet, and ROI Transformer, respectively. Our model performs well in both the single-scale and multiscale cases, with the multiscale enabling our model to be state-of-the-art. Fig. 6 shows that our model can better handle target structures with large aspect ratios such as HA, LV, GTF, and BR. Since the loss function of our SOOD model introduces the metric of IoU and uncertainty loss, the IoU metric is sensitive to the rotation angle of the target structure with a relatively large aspect ratio, the uncertainty loss can mine reliable information about the target structure. The RAE mitigates boundary discontinuity in targets with large aspect ratios and enables more accurate angle estimation. Therefore, the accuracy improvement is higher for target structures with larger aspect ratios.

2) Results on DOTA-v1.5: On the challenging DOTA-v1.5 dataset, our model achieves 70.39% and 74.10% on the single-scale and multiscale versions of the dataset, respectively. The multiscale version dataset is obtained by scaling the original satellite images by different multiples and then cropping them. This operation acts as data augmentation. As shown in Table V, the accuracy of this model is higher than other models in the categories of PL, BD, BR, GIF, SBF, and SP under the single-scale version dataset. In the single-scale version of the dataset, the SOOD model outperforms other models. The DOTA-v1.5 is similar to the DOTA-v1.0 dataset, with DOTA-v1.5 having more small targets labeled. The

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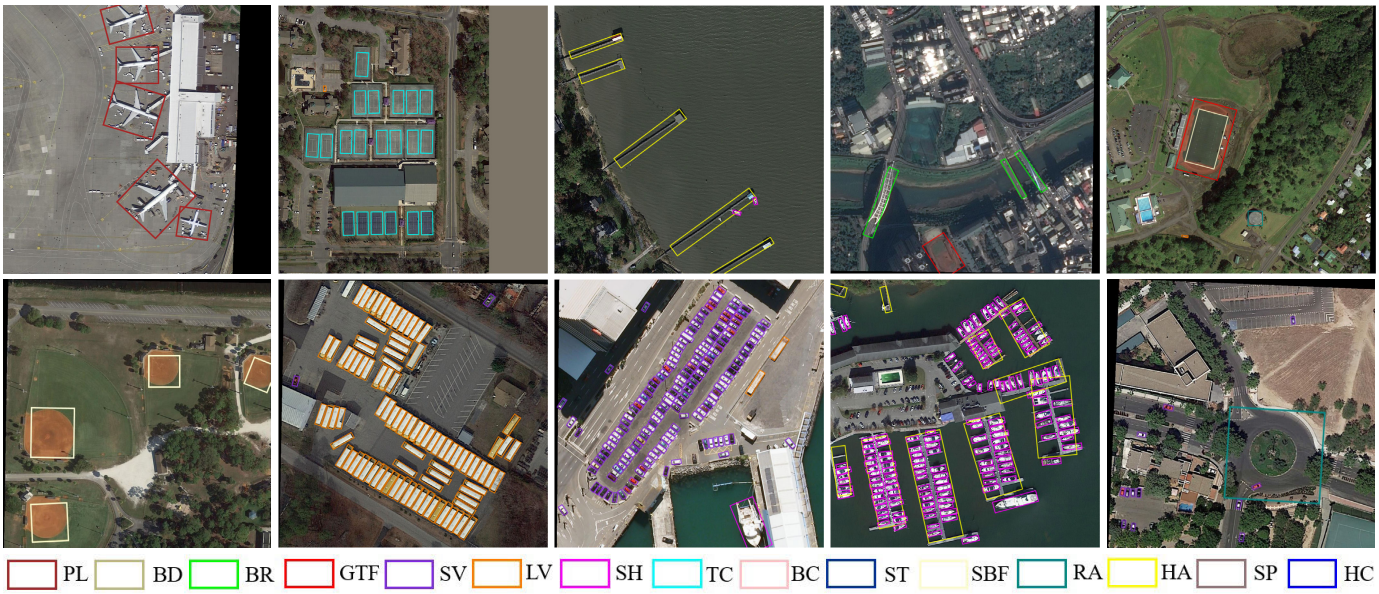


Fig. 6. Visualization of detection results on the DOTA-v1.0 single-scale dataset with our method.

TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.5 DATASET

Methods	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP
two-stage																	
Faster R-CNN-O [34]	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
Mask R-CNN [52]	76.84	73.51	49.90	57.80	51.31	71.34	79.75	90.46	74.21	66.07	46.21	70.61	63.07	64.46	57.81	9.42	62.67
HTC [53]	77.80	73.67	51.40	63.99	51.54	73.31	80.31	90.48	75.12	67.34	48.51	70.63	64.84	64.48	55.87	5.15	63.40
ReDet [54]	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	**72.03**	73.36	70.55	63.33	11.53	66.86
one-stage																	
RetinaNet-O [27]	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
RTMDet-R-tiny [51]	77.79	83.03	48.45	73.37	59.33	**81.30**	88.89	**90.88**	**80.73**	76.26	51.81	71.59	**75.81**	75.19	54.36	20.01	69.30
AO^2-DETR [55]	79.55	78.14	42.41	61.23	55.34	74.50	79.57	90.64	74.76	77.58	53.56	66.91	58.56	73.11	69.64	24.71	66.26
FSDet [56]	**80.81**	81.22	51.35	72.89	54.53	74.12	81.07	90.81	78.40	76.85	52.81	70.42	73.73	71.89	68.45	14.67	68.37
SOOD	80.32	84.41	52.59	74.77	58.48	76.90	86.97	90.87	78.62	76.56	62.93	71.16	74.64	76.04	55.97	25.09	70.39
SOOD (Multiscale)	80.76	**84.89**	**56.31**	**78.81**	**71.36**	80.42	**89.37**	80.85	77.74	**84.33**	**64.99**	69.45	75.20	**76.66**	**72.60**	**31.85**	**74.10**

TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR-R DATASET

Methods	APL	APO	BF	BC	BR	CH	ESA	ETS	DAM	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
two-stage																					
RoI Transformer [12]	63.34	37.88	71.78	**87.53**	40.68	72.60	68.09	**78.71**	26.86	68.96	82.74	**47.71**	55.61	81.21	**78.23**	70.26	81.61	54.86	43.27	**65.52**	63.87
Gliding Vertex [30]	65.35	28.87	74.96	81.33	33.88	**74.31**	64.70	70.72	19.58	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
Faster RCNN-O [34]	62.79	26.80	71.72	80.91	34.20	72.57	65.75	66.45	18.95	66.63	79.24	34.95	48.79	81.14	64.34	**71.21**	81.44	47.31	50.46	65.21	59.54
one-stage																					
FCOS-O [24]	53.79	40.85	71.59	81.05	32.18	72.48	76.36	60.18	26.37	69.17	75.71	35.72	49.05	79.54	65.58	61.65	81.18	53.00	39.71	63.90	59.45
RetinaNet-O [27]	61.49	28.52	73.57	81.17	23.98	72.54	58.20	72.39	19.94	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
FOCSF [49]	61.72	**48.15**	74.40	81.03	36.16	73.80	**83.07**	63.59	**34.03**	77.52	**83.07**	39.95	55.33	80.16	70.38	67.60	81.41	**61.23**	42.42	64.65	63.86
TIOE-Det [57]	**68.65**	28.26	**76.68**	84.76	39.32	72.35	72.66	63.87	20.36	75.19	77.41	40.63	47.48	**82.61**	72.58	70.33	**81.93**	47.86	**52.06**	64.24	61.98
SOOD	62.61	46.82	72.00	81.28	**41.40**	72.56	78.63	68.91	27.53	**78.64**	81.35	46.88	**56.32**	80.57	73.74	68.36	81.43	56.91	47.33	**65.52**	**64.41**

SOOD improves on small and compact target structures like SV and SH. Our SOOD model, guided by the SALA strategy, enables the network to adaptively select better quality anchor boxes to be assigned to small targets, improving the localization of small and compact target structures.

3) Results on DIOR-R: As shown in Table VI, the SOOD was compared against eight other models on the DIOR-R dataset, and the SOOD model achieved good results. Our model outperforms other models in the BR, GF, and OP categories. Compared with the AOPG model with the same mAP, the SOOD has excellent detection accuracy on large targets such as APO, GF, and HA. Fig. 7 exhibits the detection results of the SOOD model on the DIOR-R dataset. The DIOR-R dataset is similar to DOTA-v1.0 but has more categories. As shown in Fig. 7, the SOOD is more precise in detecting target structures with larger aspect ratios such as BR, TC, and HA. Its analysis is the same as DOTA-v1.0.

4) Results on MAR20: To further demonstrate the generality of our model, we conduct comparative experiments between the SOOD model and other models on the MAR20 dataset in Table VII. In one stage, our model exceeded FCOS-O, RetinaNet-O, and S^2A-Net in mAP metrics by 15.08%, 12.34%, and 4.67%, respectively. The RAE has better detection accuracy and orientation estimation capability than

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TABLE VII
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE MAR20 DATASET

Methods	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14	A15	A16	A17	A18	A19	A20	mAP
two-stage																					
Oriented RCNN [11]	86.05	81.73	88.08	69.57	75.61	89.92	**90.49**	89.54	89.78	**90.91**	87.62	88.39	67.52	88.50	46.33	88.27	90.59	70.50	78.72	80.25	81.92
RoI Transformer [12]	85.40	81.53	87.61	78.33	**80.45**	90.49	90.24	87.58	87.93	90.89	85.88	89.29	67.24	88.20	47.85	89.11	90.46	74.59	81.30	80.00	82.72
Gliding Vertex [30]	85.85	81.53	86.80	76.35	72.22	89.90	89.84	89.38	89.14	90.77	86.20	87.45	64.94	88.28	47.01	87.84	90.40	64.94	83.90	76.83	81.48
Faster RCNN-O [34]	85.01	81.63	87.47	70.68	79.63	90.58	89.71	89.82	90.40	90.89	85.54	88.08	68.39	88.27	42.44	88.86	90.45	62.23	78.25	77.71	81.35
Double Head-O [58]	86.35	80.85	**88.90**	**82.54**	76.04	90.06	89.76	87.28	89.20	90.78	84.35	86.18	65.76	87.42	44.13	87.49	90.25	56.40	82.82	76.69	81.16
one-stage																					
FCOS-O [24]	68.50	79.70	61.00	52.34	64.00	83.30	72.90	82.28	81.11	84.55	67.70	78.77	60.86	81.26	32.82	81.84	90.60	51.06	68.12	71.10	70.69
RetinaNet-O [27]	79.04	84.31	71.05	54.72	73.18	86.59	75.57	85.51	88.65	85.84	68.20	73.22	63.51	79.72	24.10	84.85	90.32	49.19	74.96	76.07	73.43
S^2A-Net [50]	82.62	81.59	86.21	80.75	76.86	90.00	84.73	85.70	88.70	90.84	81.67	86.09	69.59	85.25	47.69	88.10	90.20	61.98	83.59	79.84	81.10
SOOD-GWD [59]	86.17	**89.06**	87.12	80.36	76.44	89.82	89.07	84.53	87.80	90.33	87.88	89.72	66.53	87.17	68.20	**89.56**	90.55	83.38	**86.40**	80.61	84.54
SOOD-CSL [58]	86.49	**89.06**	82.90	80.72	74.39	**90.69**	90.22	86.89	**90.47**	90.66	**88.85**	87.30	59.76	**89.25**	67.57	86.70	90.62	85.00	84.37	76.00	83.90
SOOD-PSC [60]	**86.75**	85.84	86.33	79.22	77.36	89.17	87.82	**90.10**	90.33	90.89	83.94	84.84	61.90	87.60	75.02	86.98	90.54	**86.82**	81.41	68.20	83.55
SOOD	86.58	85.54	86.46	79.52	79.20	89.77	88.81	88.66	90.17	90.73	82.67	**89.78**	**75.35**	88.02	**80.10**	88.06	**90.63**	84.76	84.27	**86.34**	**85.77**

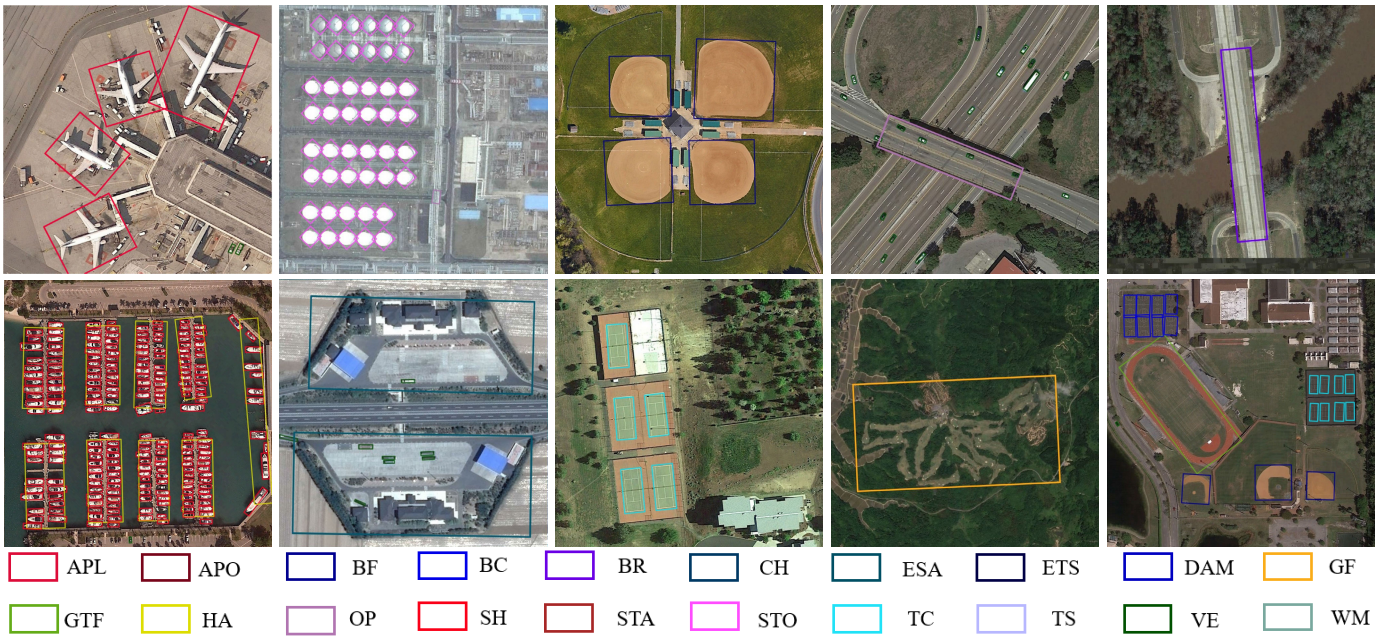


Fig. 7. Visualization of results using our SOOD on the DIOR-R dataset. Different categories of detection boxes are rendered with different colors.

TABLE VIII
INFERENCE SPEED ON THE MAR20 DATASET

Methods	Params(M)	FLOPs(G)	FPS
two-stage			
Faster RCNN-O [34]	41.14	134.40	35.37
Gliding Vertex [30]	41.15	134.40	32.19
RoI Transformer [12]	55.16	148.42	26.96
Oriented RCNN [11]	41.14	134.48	22.42
one-stage			
FCOS-O [24]	**31.93**	126.56	**62.77**
S^2ANet [50]	36.26	**121.09**	57.06
RetinaNet-O [27]	36.52	133.35	46.50
SOOD	38.99	132.36	35.87

other angle processing methods CSL [58], GWD [59], and PSC [60]. In two stages, our one-stage model can already reach the accuracy of the two-stage model and is 3.05% higher than the best ROI Transformer. In Table VIII, we compare the number of parameters, computational complexity, and inference speed of the one-stage and two-stage models respectively. The experimental platform uses the i9-13900KF CPU and NVIDIA GeForce RTX 3090 GPU and adopts the mmrotate 0.3.4 framework for the experiment. From Table VIII, the one-stage methods outperform the two-stage methods on metrics such as model capacity, complexity, and inference speed. Among the one-stage approaches, the FCOS-O model has the smallest capacity and the fastest inference speed, followed by the S^2ANet. They are superior to the SOOD model. The main reason for this is that the FCOS-O model adopts an anchor-free mechanism to avoid a large number of anchors being laid and a smaller backbone network. Compared to the FCOS-O and S^2ANet models, the accuracy comparison results in Table VII show that the overall performance of the SOOD model is improved by 15.08% and 4.67%, respectively. Hence, the SOOD model achieves a better balance between accuracy and speed of inference. In the future, it is also possible to further improve the inference speed and reduce the model size of the SOOD model by model lightweight, knowledge distillation, etc. Fig. 8 shows the comparison results of different models and different angle processing methods on the MAR20 dataset. The RAE model has superior angle prediction capability

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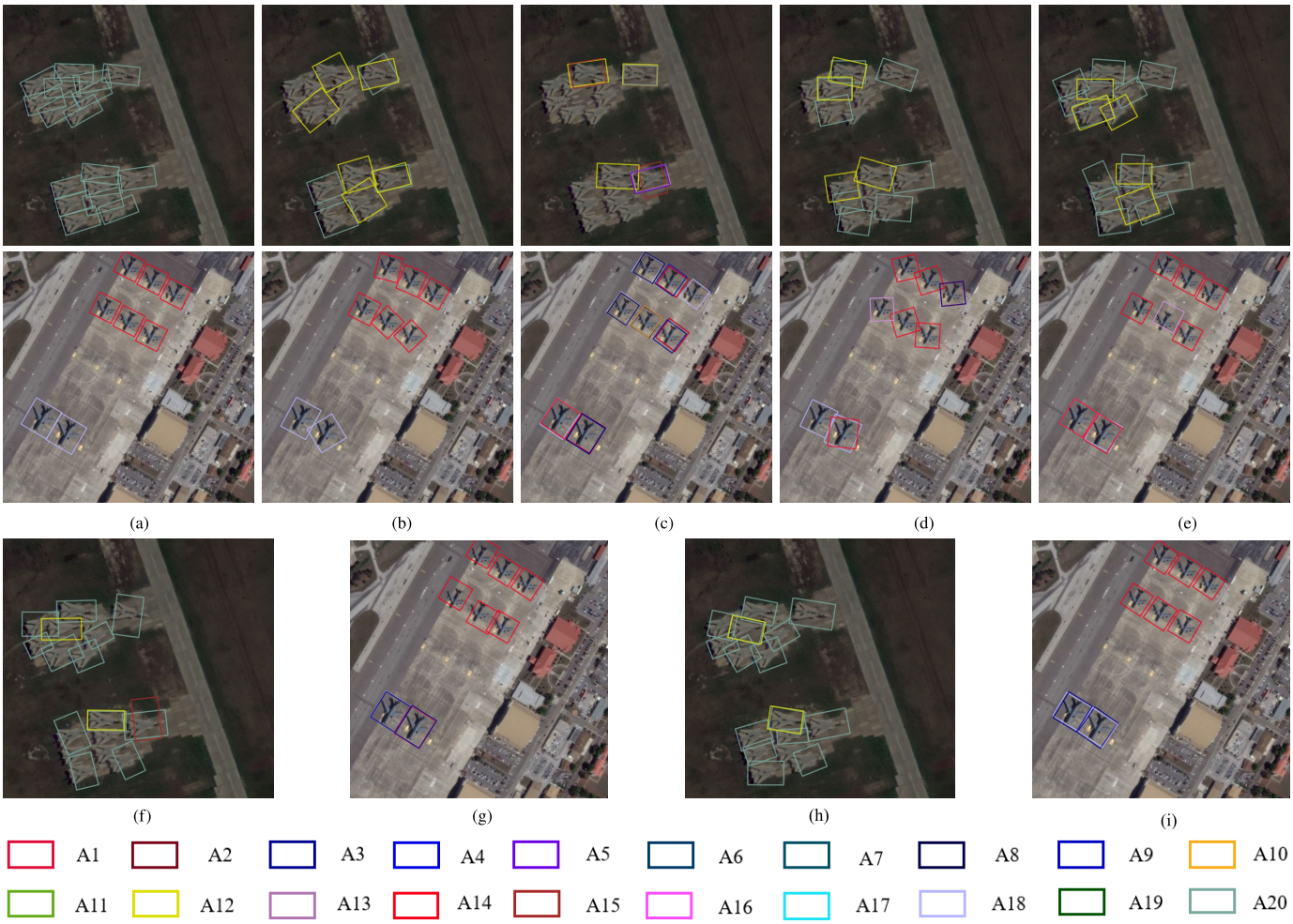


Fig. 8. Visualization of the detection results of different models on the MAR20 dataset. (a) Ground truth. (b) Oriented RCNN [11]. (c) RetinaNet-O [27]. (d) GWD [59]. (e) CSL [58]. (f) and (g) are PSC [60]. (h) and (i) are RAE. (d)–(i) are Embeddings of different angle processing methods based on the SOOD model.

compared to ORCNN [11], GWD [59], and the centroid of the prediction box is not offset compared to CSL [58], PSC [60]. It outperforms other approaches in complex and dense scenes and the estimation of object orientation (e.g., CSL [58], GWD [59], PSC [60]). Among the RetinaNet-O model detection results, there are two boxes of different colors overlapping each other. The reason is that before NMS, all the prediction boxes are scored, it is possible to encounter two or more boxes that have different colors but high confidence scores (The confusion is easily generated due to the high similarity of aircraft types A12 and A20 in the first image, and the strong likeness of aircraft in categories A18, A4, and A14 in the second image, resulting in high confidence scores on each class.). Hence it causes two or more different color boxes to overlap after NMS.

IV. Conclusion

In this article, we propose a novel one-stage oriented detector SOOD. The main contribution of SOOD is to achieve angle adaption and structural adaption of rotating targets in remote sensing images. The RAE achieves angle adaption by establishing an angle coordinate system and angle periodicity correction to alleviate boundary discontinuity and square-like problems. The SALA and SACE strategies enable precise prediction of the position and structure information of rotating targets. The SALA strategy implements the anchor structure-adaptive assignment using the internal ellipse of the rotating target. The SACE achieves structural adaption of the prediction box through a trade-off between classification and localization predictions. In addition, we design a loss function that takes into account the correlation and certainty of the predictor variables and adaptively learns reliable structural information about the rotation target. We conducted comprehensive experiments on the DOTA-v1.0, DOTA-v1.5, DIOR-R, and MAR20 datasets and the results demonstrate the effectiveness of SOOD. In the future, one focus of our work is to explore lightweight models based on this work, to better balance the efficiency and accuracy. In addition, we would like to revise the proposed model to develop an effective weakly-supervised OOD method, to further reduce the reliance on accurately labeled data and improve model robustness.

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